Moon reading light v0.2 (main board v0.1, LED board v0.2): deliverable pack

This folder holds everything needed to manufacture, print, flash, assemble and test the lamp, in build order. It is **generated** by `python3 tools/build_deliverable.py` from commit `e71b476`. Don't edit it by hand: change the design, then re-run the generator. `MANIFEST.txt` lists every file with its SHA-256 and its source in the repo.

Nothing has been ordered. Matt places every order and starts every print. The design has been through three verification rounds; it has not been built or tested on hardware yet. **What changed in v0.2** (the 18 mm LED board with the TI LP5012 driver, one RGB LED, the slim moon head, and a corrected optics model): `00-overview/what-changed-in-v0.2.pdf`.

Build order

Step	Folder	What you do	Instructions
0	`00-overview/`	Read the plan and what changed in v0.2, look at the renders and `fitment/` views, check the open decisions	`00-START-HERE.pdf`, `PLAN-…pdf`
1	`01-pcb-manufacture/`	Upload each board's Gerber zip to PCBWay (or JLCPCB) with the specs in its `FAB_NOTES`; decide how the LED board's WQFN side is assembled	`handoff-01-pcb-order.pdf`
2	`02-parts-order/`	Paste `farnell_quickpaste.txt` into Farnell's Quick Paste; the LP5012 from DigiKey UK (`digikey_order.csv`); buy the off-board items	`handoff-02-parts-order.pdf`
3	`03-3d-printing/`	Print the coupons first, then the moon head, the pack and the jig	`handoff-03-printing.pdf`
4	`04-firmware/`	Set the fuses once, then flash `release/` (or `debug/` for bring-up)	`handoff-04-firmware.pdf`
5	`05-assembly-and-bring-up/`	Assemble the boards, cable, moon and pack; first light	`handoff-05-assembly-bringup.pdf`
6	`06-test-and-verification/`	Bench checklist: electrical, firmware, thermal, optical, fit	`VERIFY-bench-checklist.pdf`
7	`07-design-reference/`	Background: electronics notes, datasheet specs, research	—

Status of the files in this pack

Board	ERC	DRC errors	DRC warnings	Unconnected	Schematic parity
main-board	0	0	2	0	0
led-board	0	0	0	0	0

The main board's 2 DRC warnings are intended: the slide-switch outline crosses the board edge where its knob overhangs. Every printed part passes the 3dprint-contract checks (12 parts). The firmware's host tests pass, and all five builds compile.

Decisions still open for Matt

See `00-overview/verification-round3-and-open-decisions.pdf`, last section. In short:

1. LED board assembly: the LP5012 is a 0.4 mm-pitch WQFN. Hand assembly needs a stencil and a hot plate; the recommendation is fab assembly of that board's bottom face (`02-parts-order/assembly-routes-cost-comparison.pdf`).
2. Green/blue headroom: keep R1 2.94 kΩ (recommended; decide after VERIFY E9 measures the received dies) or 3.65 kΩ (20 mA on every channel, −20 % white).
3. LP5012 VCAP/RIREF ground returns run ~22–26 mm: order as is and scope it (VERIFY E18, recommended) or re-lay the board centre first.
4. The LED board is not keyed and may turn ±30° with the moon (recommended: accept).
5. Carried over: reading brightness 20–30 lux (a downward reading LED in the next revision); no cell-temperature sensing on the charger, so charge at room temperature (an NTC charger in the next revision); soft-orange night mode includes some warm white (recommended: keep it).

Safety rules that apply to every step

- Meter the battery lead before plugging it in: + goes to J2 pin 1 (silk "+").
- Never power the board from a programmer or USB-UART: J4 pin 1 is a sense line only.
- Charge at room temperature, and stop using a pouch that swells.
- Do not use anaerobic thread-locker on the printed parts (it crazes PETG).